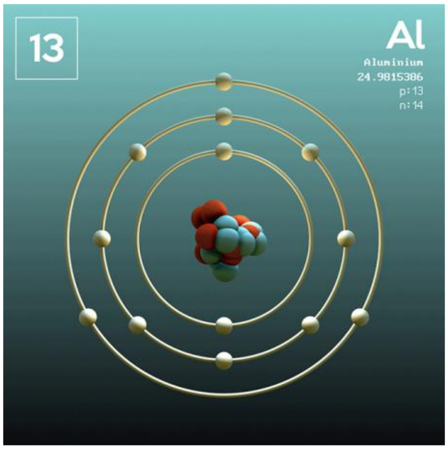
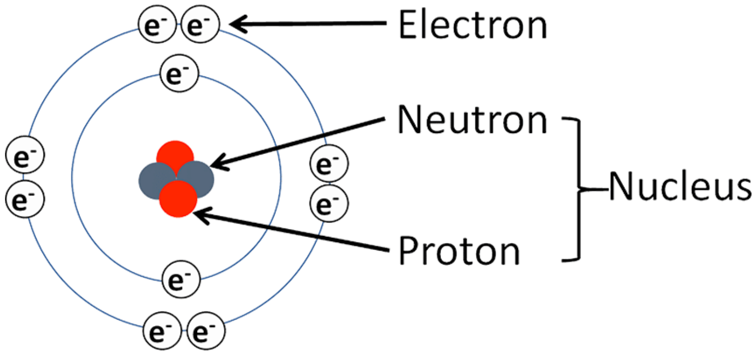
Energy Bands in Crystals

Atomic Modals

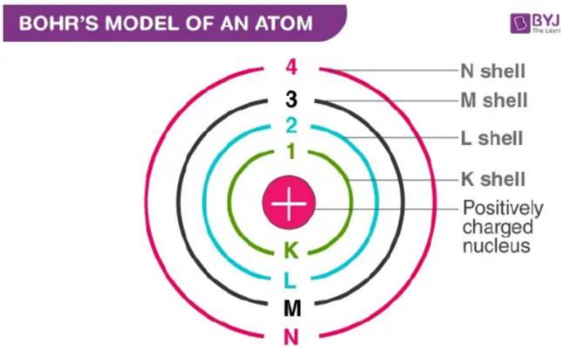
- Bohr's atom model was proposed by Niels Bohr in 1915; it is not completely correct, but it has many features that are approximately correct and it is sufficient.





Bohr's Modal

- An atom is consist of a positively charged nucleus around which negatively charged electrons revolve in different circular orbits.
- The electrons can revolve around the nucleus only in certain permitted orbits, i.e. orbits of certain radii are allowed.
- The electron in each permitted orbit has certain fixed amount of energy. The larger orbit (large radius), the greater is the energy of electron.
- If an electron is given additional energy, it is lifted to the higher orbit. The atom is said to be in a state of excitation. This state does not last long, because the electro soon falls back to the original lower orbit. As it falls, it gives back the acquired energy in the form of heat, light or other radiation.
- The protons and neutrons occupy a dense central region called the nucleus.
- Electrons orbit the nucleus much like planets orbiting the sun.
- The energy of the electron is the Bohr atom is restricted to certain discrete values, i.e. the energy is quantized. This means that only certain orbits with certain radii are allowed. Orbits in between simply does not exist.
- According to the modal, the electrons in an atom revolve round the nucleus in different orbits. Each orbit is called a shell.
- Figure shows the shells around the nucleus and its designations. These shells are filled in sequence; K is filled first, then L, M, N and so on. The maximum number of electron that each shell can accommodate is shown in the table.



Orbit Shell Designation	Maximum Number of Electrons
K	2
L	8
M	18
N	32

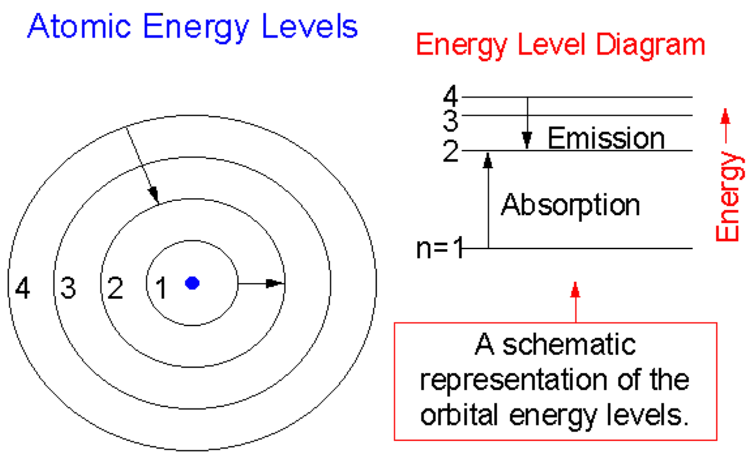
Composition of the atoms of the first eighteen elements :-

Name of element	Symbol	Atomic Number	Number of Protons	Number of Neutrons	Number of Electrons	Distribution Of Electrons K	L	M	N	Valency
Hydrogen	H	1	1	-	1	1	-	-	-	1
Helium	He	2	2	2	2	2	-	-	-	0
Lithium	Li	3	3	4	3	2	1	-	-	1
Beryllium	Be	4	4	5	4	2	2	-	-	2
Boron	B	5	5	6	5	2	3	-	-	3
Carbon	C	6	6	6	6	2	4	-	-	4
Nitrogen	N	7	7	7	7	2	5	-	-	3
Oxygen	O	8	8	8	8	2	6	-	-	2
Fluorine	F	9	9	10	9	2	7	-	-	1
Neon	Ne	10	10	10	10	2	8	-	-	0
Sodium	Na	11	11	12	11	2	8	1	-	1
Magnesium	Mg	12	12	12	12	2	8	2	-	2
Aluminium	Al	13	13	14	13	2	8	3	-	3
Silicon	Si	14	14	14	14	2	8	4	-	4
Phosphorus	P	15	15	16	15	2	8	5	-	3,5
Sulphur	S	16	16	16	16	2	8	6	-	2
Chlorine	Cl	17	17	18	17	2	8	7	-	1
Argon	Ar	18	18	22	18	2	8	8	-	0

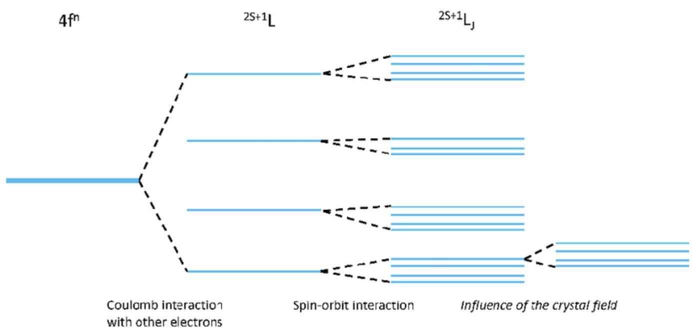
- Those outermost orbit electrons in an atom are known as "*Valance electrons*" and play an important role in determine the physical and chemical properties of material.
- The valance electrons of different materials possess different energies. Higher the energy of a valance electrons, the looser it is bound to the nucleus. I metal, the valance electrons possess very high energy that they are loosely attached to the nucleus. This makes them good conductors of electricity.
- The loosely attached valance electrons that moves randomly within the material are called "*Free electrons*". The free electrons can be easily remove or detached by applying a small amount of external energy. They server as the charge carriers in solids and determine the electrical conductivity of a material.

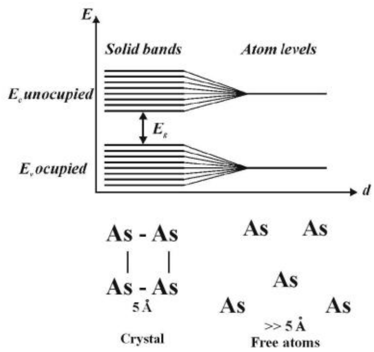
Energy Levels and Energy Bands

- According to the Bohr's modal of atom, electrons revolve around the nucleus in different orbits with different energy values. For an isolated atom these energies can be represent as follows.

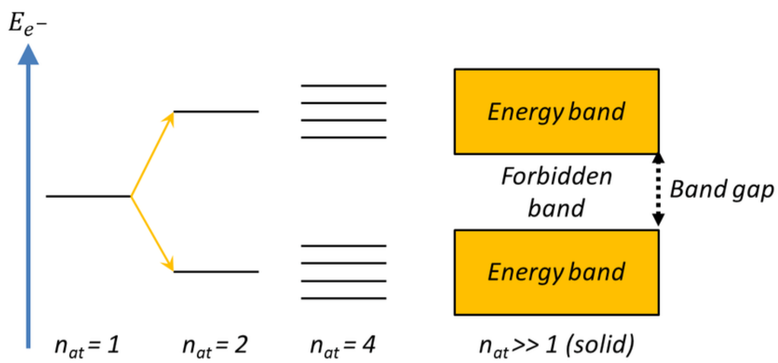


- How ever in solids large number of atoms are closely packed and atoms are not free from the influence of the neighboring atoms, Therefore energy levels are broadened and become band of energy called the *energy bands*. Each energy band consist of a very large number of very closely spaced energy levels.



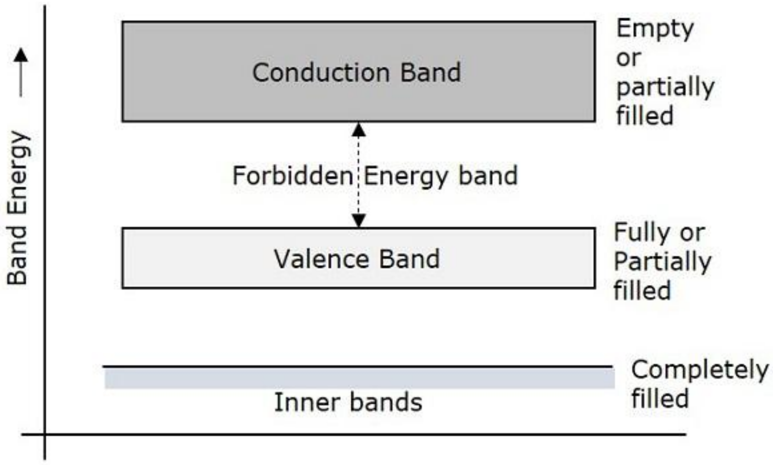


- Adjacent bands being separated by an *energy gap*referred to as *forbidden energy gap*or *forbidden gap*, which represents a range of energies that no electron can possess.

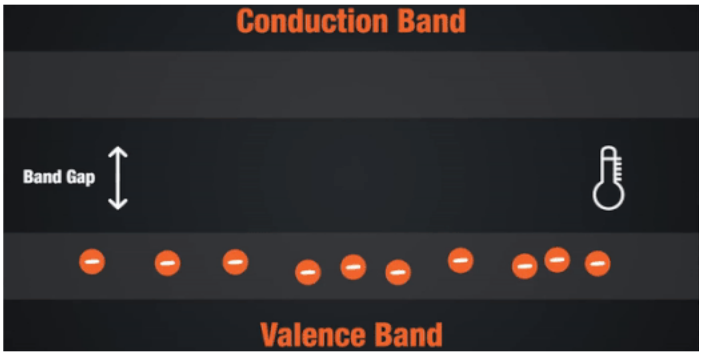


- The electrons in the outer most shell of an atom is called the valance electrons. The energy band occupancy, it may be either completely filled with electrons but can never be empty.

- The electrons which have left the valance band are called the conduction electrons. They practically leave the atom or are only weakly held to the nucleus. The band occupied by these electrons is called conduction band. This band is next to the valance band.



- If an electron in the valance band gets sufficient energy (either through electrical or thermal), it can jump across forbidden gap and enter into the conduction band . Such electrons are then free to move from atom to atom and give rice to the current flow.
- When valance electron enters the conduction band, it leaves in the valance band an empty energy level. It is now deficient of an electron and behaves like a positive charge. Such positive vacancies are called holes.



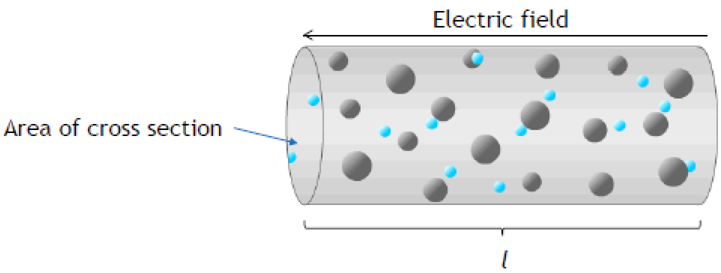
Conductivity of Materials

- Free electrons are the typical charge carries in a solid conductor. A flow of electric charge is called and electric current and it is defined as the rate of flow of charge across a cross-section of the conductor.
- If Q is the amount of electric charge moving across a cross-section of the conductor in time $t$sec, then the current $I$is given by

$$I = \frac{Q}{t}$$

- Voltage difference is applied across a conductor, and electric field will be setup in the conductor which exerts an electric force on the electrons. This cause the electrons to move and produce a current.

- Actually the electrons do not simply move in straight line along the conductor. Instead, they undergo repeated collisions with the metal ion cores, which results in a complicated motion.
- The energy transferred due to these collisions causes an increase in the temperature of metal (Joule heating).
- In spite of these collisions, the electrons drift slowly alone the conductor (in the direction opposite to the electric field) with an average velocity called *drift velocity*. It gives rice to the drift current. The speed at which they drift can be calculated.



- Consider a cylindrical conductor of length $l$and area of cross-section Aas shown in the figure.
- Let an electric field Ebe applied along the length of cylinder due to which carriers acquire a drift velocity v. Then the current I through the conductor is given by

$$I = nAve$$

- Where n is the current density (number of charge carries per unit volume) and e is the charge of the carrier.
- The current density J (current per unit area of a cross section) is given by

$$J = \frac{I}{A}nev$$

- The average drift velocity per unit electric field is called mobility of charge carrier and denoted by μ.

$$\mu = \frac{v}{E}$$

- The current density per unit applied field is called conductivity and denoted by σ.

$$\sigma = n\mu e$$

- According's to the Ohm's law,

$$I = \frac{V}{R}$$

- Also we know that the resistance R of a given conductor could be expressed in the form of

$$R = \rho\frac{l}{A}$$

- The ρ is called the resistivity. Then the conductivity σ is given by

$$\sigma = \frac{1}{\rho} = \frac{l}{AR}$$

- The reciprocal of the resistance R is known as conductance G and given by,

$$G = \frac{1}{R}$$

- If E is the applied electric field, we can write the ohms law as

$$J = \sigma E$$

- Good electrical conductors have very low resistivity (or high conductivity), and good insulators have very high resistivity (or low conductivity). However these values can change with temperature.